

ANIRUDH MOHOLKAR

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EDUCATION

University of Illinois at Urbana-Champaign | Dean's List | James Scholar | Fiddler Innovation Award

Expected 2027

Computer Engineering — Grainger College of Engineering

GPA: 3.77/4.0

Relevant Courses: Computer Systems & Programming, Data Structures & Algorithms, Distributed Systems, Artificial Intelligence, Applied Machine Learning, Reinforcement Learning

EXPERIENCE

Zebra Technologies — Software Engineer Intern, AI & Cloud Platform

May 2026 – August 2026

Selected for return offer to architect next-gen enterprise AI systems within the Zebra Data Lake

- **Agentic AI System Design:** Architected and deployed a production RAG-based conversational agent embedded in the Zebra Data Lake (ZDL), one of Zebra's first production LLM initiatives, enabling clients including Amazon, Target, and Walmart to query their own enterprise data in natural language via BigQuery and Databricks pipelines, **eliminating analyst overhead on ~85%** of routine reporting.
- **Data Pipeline & Retrieval Engineering:** Engineered a multi-source ingestion layer indexing 2TB+ of structured B2B operational telemetry into a Pinecone vector store, enabling **sub-200ms semantic search latency** across 40+ live production schemas supporting 500+ corporate users.
- **LLM Evaluation & Observability:** Independently built a proprietary prompt evaluation harness with 150+ test cases, optimizing agent **accuracy from 71% to 94%** and reducing hallucinations by 60% through iterative refinement. Work directly commended by Zebra's Chief Information Officer (CIO).

Zebra Technologies — Software Engineer Intern, Cloud & Computing

May 2025 – August 2025

- **Distributed Infrastructure Automation:** Designed and deployed an asynchronous Python and PowerShell data pipeline to extract and process system metadata, optimizing runtime from 2.5 hours to under 2 minutes: **98.5% latency reduction** across 10,000+ endpoints.
- **Backend Integration & Security Engineering:** Engineered a scalable data synchronization layer aggregating ServiceNow REST APIs and internal CMDB databases, automating vulnerability detection pipelines to isolate **3,386 security vulnerabilities** and eliminating 40 manual audit hours per cycle.
- **Agile Technical Leadership:** Appointed Scrum Master for an 8-person cross-functional engineering team, steering sprint planning, technical roadblocks, and code reviews via Jira/Git, **accelerating sprint velocity by 18%**.

PROJECTS/EXTRACURRICULARS

Design for America – UIUC Chapter | *Director of Operations, Executive Board*

May 2024 – Present

- **Technical Product Steering:** Managed SDLC and architecture across 7 teams of 56 total members, directly leading 2. Translated community requirements into high-performance MVPs, **shipping 2 to live partners for 100+ users**.
- **Operational & Financial Management:** Own **\$15K annual budget** across prototyping, software licensing, and infrastructure, strategically allocating across all 7 teams to sustain on-time delivery across 2 consecutive project cycles.

Distributed File System Development (HyDFS) | *Go, gRPC, HTTP*

Aug. 2025 – Dec. 2025

- **Collaborative System Design:** Co-engineered a fault-tolerant distributed file system via hybrid RPC/HTTP architecture and gossip-based membership protocol, achieving **99%+ node failure detection accuracy**.
- **Consistency & Fault Tolerance:** Maintained a replication factor of 3 via background merges and automated re-replication; secured metadata integrity and seamless data recovery during concurrent node crashes and rejoins.
- **Stream Processing:** Built a distributed engine with leader-driven scheduling and bidirectional RPC logging, benchmarked against Spark Streaming using **500K-user workloads** with **<100ms latency operations**.

Operating Systems Kernel Development | *C, RISC-V*

Jan. 2025 – May 2025

- **Virtual Memory & Scheduling:** Engineered a custom RISC-V operating system kernel in C, developing a virtual memory manager with dynamic page allocation and a preemptive process scheduler.
- **Concurrency & Low Latency Performance:** Optimized page allocation and address translation, **reducing memory overhead by 28%** under concurrent workloads while diagnosing and resolving low-level race conditions.

Chicagoland Help Forum | *Founder* | *React, Node.js, GitHub Actions CI/CD*

Dec. 2020 – May 2024

- **Community Impact:** Founded and scaled a community platform to 300+ users, **raising \$5,400+** for under-served students with 100% donor transparency — sustained over 3 years while enrolled full-time.
- **Technical Implementation:** Designed and maintained a responsive full-stack web platform generating **13,000+ page views** in 2024; shipped continuous feature updates via GitHub Actions CI/CD pipeline.

TECHNICAL SKILLS

AI & Agentic Systems: Agentic Workflows (ReAct/Reflection), LangChain, RAG Architecture, Vector Databases (Pinecone, Milvus), Prompt Evaluation (Evals), LLM Observability

Distributed Systems & Infrastructure: Fault-Tolerant Architecture, Microservices, Consensus Protocols (Raft/Paxos), Docker, Kubernetes, AWS (EC2, Lambda, Bedrock), CI/CD Pipelines

Low-Level & Systems: Linux Kernel Development, Multithreading/Concurrency, Memory Management, SystemVerilog, RISC-V/SLC-3 Architecture, Socket Programming

Languages & Tools: Python, C, C++, Java, JavaScript, SQL, Git, gRPC, REST APIs, Jira